

Master of Science in Engineering Portfolio

Uriah D. Villa

Master of Science in Engineering

Ira A. Fulton Schools of Engineering, Arizona State University – Polytechnic

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SUMMARY

Robotics Engineer with a 4.0 GPA in an accelerated M.S. in Engineering program at Arizona State University. Possesses hands-on internship experience in microcontroller applications, embedded systems design, and data analysis with Python at Microchip Technology. Proven ability to develop and implement solutions from motor control projects to edge-AI models for agricultural technology. Seeking a full-time role in embedded systems or robotics development.

EDUCATION

Master of Science in Engineering

Arizona State University • Tempe, AZ • GPA 4.00 • January 2025 - Anticipated December 2025

- Achievements: Accelerated Master's Program (4+1), Dean's List (Spring 2025)
- Relevant Coursework: Principles of System Engineering, Mechatronic Systems, Power Converters and Systems

Bachelor of Science in Engineering (Robotics)

Arizona State University • Tempe, AZ • GPA 3.44 • August 2020 - December 2024

- Achievements: Graduated Cum Laude, Dean's List (Spring 2021/22/24, and Fall 2024), Fulton Phoenix Award (Spring 2021)
- Relevant Coursework: Computational Modeling, Embedded Systems Design, Robotic Systems Development

SKILLS

Software Languages: C, C++, Python

IDE & Software: PSoC Creator, MPLAB X IDE, Visual Studio, KiCad, Cadence, MATLAB, Simulink

Hardware & Lab: 8/16/32-bit MCUs (PIC18/dsPIC33/PIC32), Embedded Systems Design, PCB Layout, Macro/Micro Soldering, Oscilloscopes, Digital Logic Analyzers, Power Supplies, Function Generators

Learned Concepts: Edge-AI, Motor Control, Digital & Analog Circuits, System Engineering, Robotics, and Autonomous Systems

EXPERIENCE

MCU Applications Intern

Microchip Technology Inc.

May 2025 - Present, Chandler, AZ

- Managed project workflows and tracked engineering tasks using JIRA and Confluence, ensuring alignment with team deadlines and objectives.
- Authored and maintained 5+ customer-facing application examples for PIC18 MCUs, ensuring version control and code quality using Bitbucket and SourceTree.
- Developed robust firmware solutions for customer evaluation boards, collaborating closely with cross-functional teams to facilitate seamless integration and support for PIC18 MCU products.

Support Engagement Engineer (SEE) Intern

Microchip Technology Inc.

August 2024 - May 2025, Chandler, AZ

- Automated data analysis by co-developing Python scripts to process Salesforce data, reducing reporting time by 10+ hours per week and presenting new insights into customer engagement trends with the CAE group.
- Constructed a machine learning model using Python and relevant libraries to estimate case complexity, aiding CAEs with forecasting support workflows.
- Streamlined escalation workflows by integrating Python automation tools with JIRA, enhancing coordination between the SEE and CAE teams while accelerating resolution times for high-priority cases by 20%.

Corporate Applications Engineer (CAE) Intern

Microchip Technology Inc.

August 2023 - August 2024, Chandler, AZ

- Managed and resolved 100+ technical support cases for corporate clients via Salesforce, providing application support for successfully implementing Microchip products while collaborating with other engineers.
- Designed and developed a motor control project with a BLDC motor using a 16-bit MCU (dsPIC33EP256MC506). Oscilloscopes, digital logic analyzers, power supplies, and function generators are heavily used.
- Automated diagnostic test suites were used for embedded systems, enabling consistent firmware validation and streamlining troubleshooting processes for internal engineering and customer-facing applications.

Advising Student Services

Coordinator, Senior - Junior Manager

February 2023 - Present, Mesa, AZ

Arizona State University

- Supported the Academic Advising Office by coordinating front desk operations using Salesforce and Microsoft Office Suite to optimize scheduling and student inquiries.
- Used Google Apps like Sheets and Forms to generate newer and faster ways for students to meet their advisors, cutting the time by 50%.
- Generated a QR code system to allow students to manually enter their information for Express Advising, cutting front desk resource time by 70% and increasing the number of student visitors by 20%.

PROJECTS

SolarSENSE - Edge-AI Soil Health Monitor

Team Lead | SolarSPELL & Arizona State University • January 2025 - May 2025

- Led a team to develop and demonstrate a TinyML model on an 8-bit MCU for the ASU SolarSPELL Initiative, designed to provide actionable soil health data to farmers in Rwanda.
- Collaborated with stakeholders to translate real-time sensor data into insights for small-scale agricultural applications.
- Presented the fully realized Edge-AI model at the Spring 2025 ASU Innovation Showcase.

ShadeShift - Therapeutic Eyewear for Post-Traumatic Neurological Rehabilitation

Team Lead | Barrow Neurological Institute - Dignity Health & Arizona State University • January 2025 - May 2025

- Spearheaded the design and successful integration of a 4-in-1 therapeutic device featuring customizable hues, gradable tints, stroboscopic training, and binasal occlusion technology.
- Resolved critical engineering challenges by redesigning the hardware architecture to an external enclosure due to weight constraints and transitioning from PNLC to PDLC film to achieve reliable stroboscopic effects.
- Utilized iterative 3D printing to develop custom, lightweight snap-on frames for binasal occlusion and customizable color therapies.
- Collaborated with Dignity Health to create a holistic solution for patients with photophobia and visual processing disorders following brain injuries.

Team Hephaestus - Landing System for Hypothesized Surfaces

Embedded System Design Lead | Psyche Mission • January 2024 - December 2024

- Created a tailored PCB using KiCad, incorporating PIC and PSoC microcontrollers alongside motor driver and sensor conditioning circuitry to support reliable hardware testing across hypothesized landing surfaces.
- Engineered a Hall effect sensor-based positioning system for a rover platform, achieving $\pm 2^\circ$ rotation accuracy to six destinations, and demonstrated 100% reliability over 500+ automated cycles using embedded C and PIC18 hardware.
- Wrote firmware to manage all system functions, including user input, door latch control, and LED indicators, leading to a fully functional and integrated final assembly.
- Aided in the physical testing of the lander on different slopes up to 36° , with a success criterion of $>20^\circ$.
- Project website and information: <https://psyche.ssl.berkeley.edu/landing-system-for-hypothesized-surfaces-asu-a/>

Home Room Weather Control System

Embedded System Design Lead | Arizona State University • August 2023 - December 2023

- Led hardware development, creating PCB schematics in Cadence, integrating sensors (temperature, light), motor drivers, voltage regulation, and ESP32 Wi-Fi communication.
- Developed embedded C firmware on PIC18 microcontrollers (via MPLAB X IDE) for I2C communication, sensor data acquisition, and motor control; optimized performance through register organization, modular functions, and debugging.
- Programmed ESP32 modules in Python for MQTT communication, enabling remote monitoring and control over Wi-Fi.
- Delivered a fully functional prototype and tabletop demo, with lessons learned applied to hardware reliability, testing, and PCB design best practices.
- Project website and information: <https://egr314-team-305.github.io/Team305.github.io>

CERTIFICATIONS

Universal Robots e-Series Core Track | Universal Robots • 2024

- The Core Track teaches all essential concepts, terminology, and programming commands needed to operate a UR robot.

Universal Robots e-Series Application Track | Universal Robots • 2024

- The Application Track teaches specific knowledge and skills that are applicable to manufacturing.

Universal Robots e-Series Pro Track | Universal Robots • 2024

- The Pro Track builds upon the skills learned in the core track, but teaches more complex topics.

Simulink Fundamentals | MathWorks • 2024

- Complete understanding of MathWorks Simulink Software Fundamentals.

Certificate of Arizona Technical Skills Standards | Arizona Department of Education • 2019

- This instructional program prepares students for entry-level employment, further training, and/or post-secondary education in Electrical Engineering

Introduction

This portfolio will serve as a culmination of my academic journey in the Master of Science in Engineering program, highlighting the integration of technical knowledge and professional skills I have developed. The following sections detail three of my most notable accomplishments, which I believe best represent the depth and quality of my graduate education experience at Arizona State University: the SolarSENSE soil monitoring project, the ShadeShift therapeutic eyewear system, and a techno-economic microgrid design for a critical load hospital in Hawaii.

These three projects were selected for my portfolio because they form a cohesive narrative of my growth as an engineer. These projects collectively demonstrate my progression from applying targeted technical skills to leading complex, interdisciplinary teams, and ultimately, conducting high-level, independent analysis. Each project is based on solving a real-world problem, forcing me to push beyond theoretical concepts and grapple with practical constraints, stakeholder needs, and system integration. From global development and mechatronic device innovation to critical infrastructure design, these experiences have helped me expand my engineering knowledge, technical expertise, and problem-solving skills.

The SolarSENSE project was a hands-on dive into systems integration for global development, where the primary role was to merge a TinyML model with embedded hardware to create a low-cost soil sensor for farmers in low-income-impacted countries. This project was developed in parallel with the ShadeShift project, a mechatronics challenge where, as team lead, I guided the development of a 4-in-1 therapeutic device for patients suffering from various forms of traumatic brain injury (TBI) or undergoing neurosurgical recovery. This role required me to resolve significant issues related to hardware and materials science. Ultimately, the

techno-economic microgrid analysis demonstrates my ability to manage a complex engineering problem from start to finish. This project utilized the XENDEE platform to design a resilient power system for a real-world hospital, striking a balance between economic and engineering objectives. Together, these accomplishments illustrate a comprehensive and practical engineering education.

Project 1: EGR 598 SolarSENSE Soil Monitoring for Rwandan Coffee Farmers

The goal of this project was to address the significant challenges faced by rural coffee farmers in Rwanda, who often lack access to affordable, simple, and practical tools for soil health monitoring (Pisors et al.). The primary objective was to develop a low-cost, stand-alone system that provides farmers with real-time, actionable insights to improve crop yields and optimize resource management in environments with limited connectivity (Pisors et al.). As a team, we developed an IoT-based sensor solution using the LILYGO T-Higrow ESP32 board. The system was designed to leverage Tiny Machine Learning (TinyML) to provide on-device soil quality predictions without requiring an internet connection. The overall process involved analyzing a soil dataset provided by our Rwandan partners, training a classification model using the Edge Impulse platform, and afterward deploying that lightweight model directly onto the ESP32 microcontroller ([see Appendix A](#)).

My central role in this team project was the significant technical task of system integration. I was specifically responsible for merging the exported TinyML model code from Edge Impulse with the LILYGO T-Higrow's existing firmware. This firmware was already configured to use the ESP DASH framework to host a web-based dashboard on the ESP32 itself. My work began with collaborating on the initial creation and labeling of data for our ML model.

A critical challenge emerged when we realized the initial NPK (Nitrogen, Phosphorus, Potassium) dataset did not match our board's available sensors. I then focused on pivoting our model's design, leading the effort to retrain and validate a new model that successfully used the board's onboard salinity (EC) and moisture sensors, ultimately achieving over 94% accuracy. Following this, I led the complex integration of the new model's code with the ESP DASH firmware inside the Arduino IDE. This iterative process was a demanding process where I utilized Google Gemini to understand the complex file structure and to effectively debug compilation and runtime errors, which significantly accelerated development (Villa, *Final Individual Reflection on TinyML and Soil Sensing in the SolarSENSE Project*).

The outcome was a functional, integrated system. I successfully flashed the combined code onto the LILYGO T-Higrow board, resulting in a device that could read live soil data (salinity and moisture), process it locally through the integrated TinyML model, and display the final soil health prediction (e.g., "Too Dry," "Healthy") alongside the raw sensor data on its self-hosted ESP DASH web dashboard. The SolarSENSE product provided a complete, offline "sense-and-predict" solution for the farmers ([see Appendix B](#)).

Reflection on Project 1

This project was a substantial and practical application of engineering principles, which enabled me to move beyond theoretical knowledge and gain valuable insights into the practical deployment of TinyML models on microcontrollers with limited resources. The most valuable learning experience I gained was adapting our model to the hardware available to us, which is a common real-world engineering challenge that emphasizes the challenges and satisfaction of

system integration, particularly when bridging ML outputs with embedded system firmware and user interfaces.

Through this work, I directly developed and enhanced several key technical skills. I gained more hands-on proficiency in the end-to-end TinyML pipeline, from data preparation and sorting in Edge Impulse to deployment on an ESP32. My skills in programming, working with external libraries like ESP-DASH, and debugging complex, integrated code in the Arduino IDE have grown substantially. This project also taught me to leverage AI developer tools, such as Google Gemini, for rapid code comprehension and efficient debugging. The project was naturally interdisciplinary, sitting at the crossroads of embedded systems, machine learning, and global development. It served as a powerful example of applying advanced engineering solutions to a tangible agricultural problem in a resource-constrained environment. Successfully integrating the separate software components and seeing the final, functional prototype perform as intended was a deep personal accomplishment. It validated the intensive effort required and solidified my understanding of and interest in embedded AI and its real-world applications ([see Appendix B](#)).

Project 2: EGR 555 ShadeShift: Smart Modular Eyewear for Neurological Recovery

This project, developed in collaboration with Dignity Health, addressed the clinical need for a holistic solution for patients recovering from traumatic brain injuries (TBIs) who suffer from photophobia and visual processing disorders. The primary objective was to design and prototype "ShadeShift," a 4-in-1 therapeutic device featuring customizable color hues, gradable electrochromic tints, stroboscopic training, and binasal occlusion technology, all integrated into a wearable, all-in-one modular eyewear system (Villa et al.).

As the team lead, I was responsible for guiding the project's direction and executing the core design and integration. The team's process involved iterative CAD design, 3D printing of frames, and integrating electro-optic materials with a microcontroller. I learned and applied 3D modeling and printing to develop the custom, lightweight snap-on frames that enabled the modularity for binasal occlusion and customizable color therapies.

A significant part of my contribution involved resolving some critical engineering challenges in our prototype. Initially, the weight of the onboard electronics made the glasses uncomfortable and impractical. I solved this by redesigning the hardware architecture, moving the microcontroller and battery to a compact external enclosure. Furthermore, our initial material for stroboscopic training, PNLC (Polymer Network Liquid Crystal) film, proved unreliable as the controller for the film needed to be reverse-engineered entirely for our use case. I led the transition to Polymer Dispersed Liquid Crystal (PDLC) film, which was essential to achieving the reliable and fast-switching stroboscopic effects required for visual therapy.

The outcome was a successful, fully functional prototype and proof of concept for our idea. The ShadeShift system successfully demonstrated all four therapeutic functions in a single, integrated device. The final assembly successfully demonstrated a proof-of-concept for the electrochromic lenses, which control tint, and the 3D-printed magnetic clip-ons for the colored filters, binasal occlusion, and the PDLC film for stroboscopic training, thereby validating the modular and multi-functional design (Uriah Villa, 2025) ([see Appendix C](#)).

Reflection on Project 2

This project represented a significant accomplishment in systems integration and engineering problem-solving. My role as team lead required me not only to manage the project's

goals but also to delve into materials science and hardware architecture to identify and solve fundamental design flaws. The decision to pivot from an all-in-one frame to an external enclosure, along with the research and testing that led to the transition from PNLC to PDLC film, were critical leadership and engineering decisions that saved the project and ensured its completion and success within the allotted class and project duration.

On a technical level, I gained new expertise in additive manufacturing (3D printing) and rapid prototyping, moving from a CAD concept in Tinkercad to a functional, user-tested device. This project was a definitive example of mechatronics, forcing me to integrate electronics (microcontrollers, PWM signals, and analog signals), optics (PDLC and electrochromic films), and mechanical design (3D-printed modular frames).

The most significant aspect of this project was its interdisciplinary nature. I served as the team's primary point of contact with Dignity Health, enabling me to collaborate directly with clinical stakeholders. This experience taught me how to translate clinical needs into concrete engineering specifications, ultimately leading to a holistic solution. It solidified my ability to bridge the gap between medical engineering, patient-centric design, and mechatronic engineering.

Project 3: EGR 586 Microgrid Design for a Critical Facility Load

This project was an independent analysis to design and optimize a microgrid for a critical facility, The Queen's Medical Center in Honolulu, Hawaii. The primary objective was to use an industry-standard software platform, XENDEE, to model a microgrid that could provide both economic savings against the utility's Time-of-Use (TOU) tariff and enhance energy resilience during grid outages of varying durations (Villa, *EGR 586 Project 1*). My process involved the

complete, independent execution of the microgrid design. I began by identifying the critical facility, researching its function as a Level I Trauma Center, and selecting an appropriate load profile for the facility. I then built the project in XENDEE, which involved defining all financial and technical parameters, including the complex local utility tariff, demand charges, and the capital costs for Solar PV, Battery Energy Storage Systems (BESS), and diesel generators.

I independently ran and analyzed multiple scenarios: a "Best Case" for economic optimization, a "Low Asset Cost" case where each asset was the lowest possible cost according to the Lazard Report, a "100% Renewable" mandate, and a series of resilience scenarios for 1-day, 3-day, and 2-week outages. My initial analysis revealed that the "Best Case" economic design was surprisingly robust, capable of handling a 2-week outage with no added cost. Skeptical of this "perfect" result, I conducted a second analysis with more realistic constraints on solar and battery physical space, which produced a more nuanced and practical design that incorporated the diesel generator for extended outages ([see Appendix D](#)).

Reflection on Project 3

This project was a significant demonstration of my independent analytical abilities. It directly reflects my growth in engineering knowledge by requiring me to understand and model the complex interplay between economic optimization (NPV, LCOE) and engineering resilience for a critical societal asset. I learned the fundamentals of microgrid design, load-profile analysis, and the dispatch logic of hybrid systems (PV + BESS and Generator).

My technical expertise grew through my proficiency in the XENDEE platform. I moved beyond theoretical class concepts to independently manage a complex, data-driven simulation from start to finish. I learned how to interpret techno-economic reports and energy dispatch

graphs to understand why the optimizer made certain decisions, such as using solar energy to charge the battery for evening peak shaving.

Most importantly, this project showcases my academic growth and critical thinking skills. When the initial results seemed too good to be true, I did not simply accept them. I took the initiative to question the model's assumptions, impose more realistic constraints, and run a new analysis. This second, self-motivated investigation demonstrated a deeper, more practical understanding of the engineering problem and led to a more realistic and defensible microgrid design. This independent project successfully integrated my knowledge of power systems, economics, and specialized engineering software to design and optimize a microgrid for a critical load facility ([see Appendix D](#)).

Reflections

Reflecting on my first semester, my perspective on engineering growth was primarily academic in nature. I had entered the MS Engineering program for a practical reason: to expand my education and fulfill the requirements for my internship with Microchip Technology. At the time, I viewed this path as a worthwhile trade-off, equating "growth" with simply learning more complex technical topics than what I had learned as an undergraduate. My initial thought was that the program would provide a bridge to apply this theoretical knowledge to actual, real-world applications and to expand my ability to work within a proficient team.

The three projects I detailed in this portfolio illustrate how my understanding of "growth" evolved far beyond that initial, narrow definition. The common thread connecting them all is their focus on solving tangible, human-centered problems, whether it was designing an accessible soil sensor for Rwandan farmers, a therapeutic device for TBI patients, or a resilient

power grid for a critical hospital. While the “SolarSENSE” and “ShadeShift” projects honed my skills in embedded systems and mechatronics, all three demanded a high level of systems engineering. I found my growth was truly measured by my ability to perform engineering analysis: to step back, look at the entire problem, and find ways to address all the interconnected needs and available resources.

Through these experiences, I came to understand that engineering is not purely about isolated technical skills. My leadership on the 'ShadeShift' project, in particular, was a profound lesson in this. Communicating with stakeholders at Dignity Health, managing team delegation, and making critical design pivots based on real-world constraints were as integral to the project's success as any circuit design. This program taught me that engineering is fundamentally about finding the right solution to a problem. That solution often requires a multifaceted toolkit of communication, project management, and interdisciplinary cooperation, not just technical proficiency in specific topic areas.

As I look to the future, my career goals have been reshaped by this perspective. I am no longer just seeking a job; I am seeking to be part of a multifaceted team where I can work with people from diverse backgrounds to solve problems we are passionate about. The field of Systems Engineering now seems like the natural next step, and this program has given me both the drive and the foundational experience to pursue it. This MS program provided far more than an advanced degree; it offered a new perspective on my role as an engineer and solidified my desire to work on real-world problems I can be proud of.

References

Pisors, Jacob, et al. Team Final Report: Soil Monitoring for Rwandan Coffee Farmers. Arizona State University, 4 May 2025.

[Uriah Villa]. *EGR 555 ShadeShift Final Project Demo* [Video]. YouTube.com. 1 May, 2025
<https://www.youtube.com/watch?v=Cl6hf7lQr3M&t=415s>

Villa, Uriah D. EGR 586 Project 1. Arizona State University, 17 Oct. 2025.

---. Final Individual Reflection on TinyML and Soil Sensing in the SolarSENSE Project. Arizona State University, 6 May 2025.

---. ShadeShift: Smart Modular Eyewear for Neurological Recovery and Visual Stress Alleviation. Arizona State University, 2 May 2025.

Appendix A: EGR 598 Team Final Report

 Team Final Report: Soil Monitoring for Rwandan Coffee Farmers

Appendix B: EGR 598 Final Individual Reflection

 Final Individual Reflection on TinyML and Soil Sensing in the SolarSENSE Project

Appendix C: EGR 555 ShadeShift Team Final Report

 ShadeShift: Smart Modular Eyewear for Neurological Recovery and Visual Stress Alleviation

Appendix D: EGR 586 Project 1 Report

 EGR 586 Project 1